Prodigy Anesthesia Quick Review

Burns

What are the four types of burns?	Chemical, electrical, thermal, and inhalation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 901.*
How are burns classified?	Burns are classified as either first, second, third, or fourth-degree. First-degree consists of erythema with only microscopic damage to the superficial epidermis. Second-degree burns, also called partial thickness, extend through the epidermis into the dermis. Spontaneous regeneration of the skin is possible with second-degree burns. In third-degree burns, total destruction of the skin, dermal appendages, and epithelial elements occurs with no spontaneous regeneration of the skin possible. Fourth-degree burns involve muscle, fascia, and bone. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 1204.*
How does the rule of nines allocate the surface area percentages according to arms, legs, head, and trunk in adults?	In adults, each arm represents 9%, each leg represents 18%, the entire trunk is 36%, the head is 9%, and the perineum is 1%. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 1204.*
What is the initial treatment for chemical burns and why?	Chemical burns are caused by disruption of the cellular components of the skin by the chemical irritant. The initial treatment is irrigation with water or saline irrigation because the chemical will continue to damage tissue until it is removed. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 901.*
What are the characteristics of electrical burns?	Electrical burns vary in severity based on the voltage and duration of contact with the source. Electrical burns will cause damage at a point of entry and exit. These wounds may appear superficial and conceal internal damage to nerves, vessels, muscle and bone. Significant damage to muscle tissue can cause the massive release of myoglobin which can place these patients at risk for renal damage. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 901.*

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Why is fluid resuscitation such a critical part of the care of a burn patient?	The area of edema from the burn acts as a 'third-space' into which fluid becomes sequestered. Fluid administration must be sufficient to maintain tissue perfusion and adequate urine output. Varying formulas are used to calculate the appropriate amount of fluid with which to resuscitate the patient. The most commonly used formula for determining the amount of fluid to administer in the first 24 hours is % body surface area X Kg X 2 to 4. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 1208.*
What chemical mediators are released with burn injury and what responses would you expect to see as a result?	Burned tissues release histamine, prostaglandins, bradykinin, nitric oxide, serotonin, and substance P. In small burns, the mediators are responsible for local inflammation. Severe burns can result in the widespread release of these mediators and a systemic inflammatory response syndrome can result. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 1205-1206.*
What are the adverse effects of carbon monoxide that occur with smoke inhalation?	Carbon monoxide has an affinity for hemoglobin that is 200 times greater than that of oxygen, impairs mitchondrial function, uncouples oxidative phosphorylation and reduces ATP production resulting in metabolic acidosis, and shifts the oxyhemoglobin dissociation curve to the left, impairing the unloading of oxygen to the tissues. In addition, it acts as a direct myocardial toxin and can prevent survival in resuscitation efforts during cardiac arrest. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1515-1516.*
Is the SpO2 a good indicator of oxygenation following a severe burn?	No. The PaO2 and SaO2 may be normal despite high levels of carboxyhemoglobin (carbon monoxide poisoning). *Lee CY. Manual of Anesthesiology. Singapore: McGraw-Hill; 2006: 326-327,333.*
What is the difference between upper and lower airway inhalation injury?	Upper airway inhalation injury is usually caused by superheated air and steam. The larynx, epiglottis, tongue, and pharyngeal tissue swell rapidly and can result in complete airway obstruction. Lower airway inhalation injury is usually due to exposure to soot and chemical toxins. The toxins produce acidic and alkaline compounds that disrupt the capillary permeability of the pulmonary epithelium and can result in alveolar damage. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 901-902.*

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What is the minimum urinary output for a patient with high-voltage electrical burns?	In patients with high-voltage electrical burns, the minimum urinary output to maintain is 1-1.5 mL/kg/hour. In ordinary burns in adults, the minimum urinary output is 0.5 mL/kg/hour. In pediatric patients less than 30 Kg, the minimum is 1 mL/kg/hour. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 905.*
A patient involved in an MVA has multiple fractures and burns of the face and neck. The patient is breathing spontaneously, but you suspect he may have an inhalation injury. How should you approach airway management?	Establishment of a surgical airway should be the last resort due to the high risk for complications. Direct laryngoscopy or awake fiberoptic intubation are preferred, if possible. *Lee CY. Manual of Anesthesiology. Singapore: McGraw-Hill; 2006: 326-327,333.*
Is succinylcholine contraindicated for a patient with an acute burn?	No. Although succinylcholine should be avoided 72 hours after a burn injury is sustained because of the risk of hyperkalemia, it is not contraindicated in the immediate resuscitation stage. *Lee CY. Manual of Anesthesiology. Singapore: McGraw-Hill; 2006: 326-327,333.*
What is hypermetabolic phase of a burn?	The hypermetabolic phase occurs at least 48 hours after the initial injury and is characterized by hyperthermia, tachypnea, tachycardiac, and increased oxygen demand. Catecholamines and corticosteroids are the primary mediators of the response, which can last up to 2 years following the injury. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 906-907.*
A patient has burns over his entire body. How do you place ECG electrodes?	In patients undergoing surgical debridement for burns, placement of the ECG leads can be challenging. In situations where the adhesive pads cannot be placed due to the damage they would cause to the burned tissue, the leads should be stapled to the patients skin or needle electrodes should be used. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 909.*

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